

Chaithanya Virupaksha

Machine Learning Engineer · PyTorch & Applied ML

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Aalyria Technologies (UK) Ltd · Machine Learning Engineer – Spacetime UK

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Dear Spacetime hiring team,

I am applying for the Machine Learning Engineer role on the Spacetime team. My MSc dissertation at Newcastle University studied a problem close to yours: deciding which nodes in an unreliable, time-varying network can and should take part, when each link has a capacity and the system has a deadline. Spacetime orchestrates networks whose links appear and disappear over time. I would like to bring the same habits of modelling and measurement to that problem at a much larger scale.

Machine learning for resource decisions over unreliable links. I built a federated-learning simulator in Python and PyTorch that models three kinds of link failure separately: independent packet loss, time-correlated burst outages using a rate-matched Gilbert-Elliott channel, and deadline misses set by each node's uplink bandwidth. I crossed these with compression schemes and four node-selection policies. The central finding was that compression changes which nodes are feasible at all. Selection can only choose among nodes whose payload fits the deadline, so with Top-k compression the accuracy spread across policies fell from 16.4 to 2.3 percentage points. To show this was a feasibility effect and not a side effect of sending less data, I swept the deadline from 1 to 8 seconds. The advantage rose from 0 to 83.7 points and returned to -0.2 , at exactly the boundaries the link-budget equation predicted.

Tests for both software and ML. The framework has 50 unit tests that check the properties my claims depend on: that the burst channel hits its target loss rate, that identical seeds produce identical channel conditions across experiments, that aggregation weights sum to one under any drop pattern, and that greedy selection starves nodes while a staleness-bounded variant does not. My per-node logging exposed a real orchestration risk. A greedy reliability policy raised update completion from 52% to 89%, but nine of twenty nodes never delivered an update, because early failures lowered their scores and they were never polled again. I reported that trade-off rather than only the accuracy gain, and implemented a staleness-bounded selection rule that the tests show prevents this starvation.

Documentation, infrastructure and communication. I audited my own work, found and fixed eight defects and documented each one in a public changelog; one of them overturned an earlier result. Every table in the dissertation is regenerated from archived runs by script. I have also worked on a cloud-native microservices platform deployed on Azure Kubernetes Service with GitHub Actions CI/CD, and co-authored its architecture covering autoscaling and recovery. Before my MSc I spent 15 months working directly with clients on ML projects, scoping requirements and explaining results to non-technical users, which is the customer-facing side of this role.

I am based in Newcastle and available for remote work across the UK, with right to work under the Graduate Route from December 2026. The code and changelog are at github.com/chai6699/federated-learning-packet-loss. I would welcome the chance to discuss how I could contribute to Spacetime.

Yours sincerely,

Chaithanya Virupaksha